

2006 Extension 1 Term 4

QUESTION ONE

MARKS

- (a) (i) Find $\int (3x+5)^8 dx$. 1
- (ii) Use the substitution $u = 3 - x^2$ to find $\int \frac{xdx}{\sqrt{3-x^2}}$. 2
- (b) Express $\sqrt{12} \sin x + 2\cos x$ in the form of $A\cos(x + \alpha)$; where $A > 0$ and $0 \leq \alpha \leq 2\pi$. 3
- (c) (i) By using one application of Simpson's Rule (i.e. 3 function values), find an approximation for the area bounded by $y = \frac{1}{x}$, $1 \leq x \leq 2$, and the x axis, leaving your answer as a fraction in simplest terms. 2
- (ii) Using integration, find an expression for the exact area in part (i). 1
- (iii) Using your answers to parts (i) and (ii) above , find an approximation for e correct to 4 decimal places. 1

QUESTION TWO

- (a) Differentiate $\tan(x^2)$. 1
- (b) (i) Differentiate $(x \sin^{-1} \frac{x}{4} + \sqrt{16-x^2})$ and simplify. 3
- (ii) Hence evaluate $\int_0^4 \sin^{-1} \frac{x}{4} dx$. 2
- (c) (i) Given that (0,0) is the only stationary point of $y = x \sin^{-1} \frac{x}{4}$, justify whether it is a minimum or maximum turning point. 2

- (ii) Hence , or otherwise, sketch the curve $y = x \sin^{-1} \frac{x}{4}$. 2

QUESTION THREE

MARKS

- (a) If $x = a + b$ and $y = a - b$,
- (i) Show that $\cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)$. 2
- (ii) Hence find the general solutions to $\cos 4\theta + \cos 3\theta + \cos 2\theta + \cos \theta = 0$. 4
- (b) The region in the first quadrant bounded by the curve $y = 2\tan^{-1} x$ and the y -axis between $y = 0$ and $y = \frac{\pi}{2}$ is rotated through one complete revolution about the y -axis. Find the exact volume of the solid of revolution so formed. 4

QUESTION FOUR

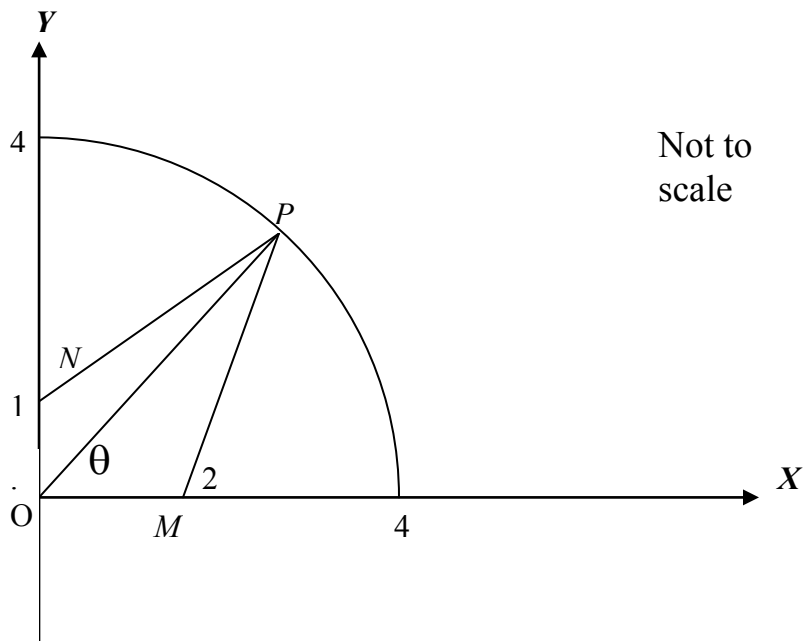
- (a) Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 x \cos^2 x \, dx$. 3
- (b) A function is defined as $f(x) = 1 - \cos \frac{x}{2}$, where $0 \leq x \leq a$. 2
- (i) Find the largest value of a for which the inverse function $f^{-1}(x)$ exists.
- (ii) Find the equation of $f^{-1}(x)$. 1
- (iii) Sketch the graph of $y = f^{-1}(x)$. 2
- (iv) Find the area enclosed between the curve $y = f^{-1}(x)$, the x -axis and $x = 2$. 2

QUESTION FIVE**MARKS**

(a) Find the value of $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x}$.

3

(b)



The diagram shows the part of the circle $x^2 + y^2 = 16$ that lies in the first quadrant. The point $P(x, y)$ is on the circle with centre at the origin O.

M is on the x -axis at $x = 2$ and N is on the y -axis at $y = 1$.

The size of angle MOP is θ radians.

(i) Show that the area, A , of the quadrilateral $OMPN$ is given by
 $A = 4\sin\theta + 2\cos\theta$.

2

(ii) Find the value of $\tan\theta$ for which A is maximum.

3

- (iii) Hence determine in surd form the coordinates of P for which A is maximum. 2

QUESTION SIX

MARKS

The straight line $y = mx + b$ meets the parabola $x^2 = 4ay$ at the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$.

- (i) Find the equation of the chord PQ and hence, or otherwise, show that $pq = \frac{-b}{a}$. 3
- (ii) Prove that $p^2 + q^2 = 4m^2 + \frac{2b}{a}$. 2
- (iii) Given that the equation of the normal to the parabola at P is $x + py = 2ap + ap^3$ and that N , the point of intersection of the normals at P and Q , has co-ordinates $[-apq(p+q), a(2 + p^2 + pq + q^2)]$, express these co-ordinates in terms of a, m, b . 2
- (iv) Suppose that the chord PQ is free to move while maintaining a fixed gradient. Find the locus of N and show that this locus is a straight line. 3

Verify that this line is a normal to the parabola.

END OF PAPER

Solutions to T4 3u 2006

1a) $\frac{(3x+5)^9}{27} + C$ #

ii) $-\frac{1}{2} \int \frac{du}{\sqrt{u}}$ $u = 3-x^2$
 $du = -2x dx$
 $= -\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}}$
 $= -\sqrt{3-x^2} + C$ #

b) $\sqrt{12} \sin x + 2 \cos x = A \cos x \cos \alpha - A \sin x \sin \alpha$
 $A = \sqrt{(\sqrt{12})^2 + 2^2} = \sqrt{16} = 4$ $A = 4$
 $-4 \sin \alpha = \sqrt{12}$ $2 = 4 \cos \alpha$
 $\sin \alpha = \frac{\sqrt{12}}{4}$ $\cos \alpha = \frac{1}{2}$ α is in 4th Quad
 $\tan \alpha = \frac{\sqrt{12}}{2} = \sqrt{3}$ $\alpha = \frac{5\pi}{3}$
 $\sqrt{12} \sin x + 2 \cos x = 4 \cos(x + \frac{5\pi}{3})$ #

c)

x	1	1 1/2	2
y = 1/x	1	2/3	1/2

Area $\div \frac{2-1}{6} [1 + 4 \times \frac{2}{3} + \frac{1}{2}] = \frac{1}{6} (\frac{3}{3} + \frac{8}{3} + \frac{1}{2})$
 $= \frac{25}{36}$

ii) Exact Area $= \int_1^2 \frac{1}{x} dx = [ln x]_1^2 = ln 2 - ln 1$
 $= ln 2$

iii) $ln 2 \div \frac{25}{36}$
 $e^{\frac{25}{36}} = 2$ $e = 2^{\frac{36}{25}} = 2.7132$ (4dp)

2a) $2x \sec(x^2)$

b) $x \cdot \frac{\frac{1}{4}}{\sqrt{1-\frac{x^2}{16}}} + \sin^{-1} \frac{x}{4} + \frac{-2x}{2\sqrt{16-x^2}}$
 $= \frac{\frac{1}{4}x}{\sqrt{16-x^2}} + \sin^{-1} \frac{x}{4} - \frac{x}{\sqrt{16-x^2}}$
 $= \sin^{-1} \frac{x}{4}$

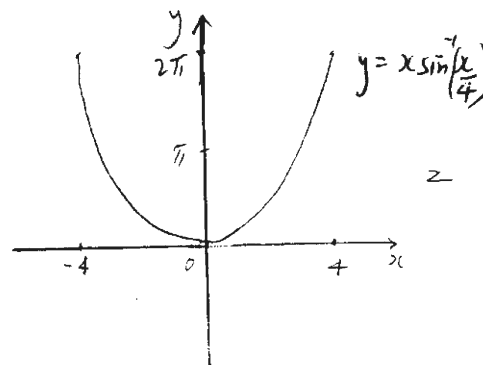
ii) $\int_0^4 \sin^{-1} \frac{x}{4} dx = [x \sin^{-1} \frac{x}{4} + \sqrt{16-x^2}]_0^4$
 $= 4 \sin^{-1} 1 + 0 - 0 - \sqrt{16} = 2\pi - 4$

c) $y' = \frac{2x}{\sqrt{16-x^2}} + \sin^{-1} \frac{x}{4}$ (from b)

x	-0.1	0	0.1
y'	-0.5	0	0.5
shape	↘	—	↗

Since y is continuous and differential for $x \leq 1$ and y' changes signs (- to +) from $x = -0.1$ to $x = 0.1$

$\therefore$ min turning point at $(0, 0)$



3a) $\cos x + \cos y = \cos(a+b) + \cos(a-b)$
 $= \cos a \cos b - \sin a \sin b + \cos a \cos b + \sin a \sin b$
 $= 2 \cos a \cos b = 2 \cos(\frac{x+y}{2}) \cos(\frac{x-y}{2})$
 since $\begin{cases} x = a+b \\ y = a-b \end{cases} \Rightarrow \begin{cases} a = \frac{x+y}{2} \\ b = \frac{x-y}{2} \end{cases}$

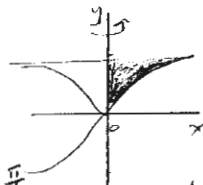
ii) $(\cos 4\theta + \cos 2\theta) + (\cos 3\theta + \cos \theta) = 0$
 $2 \cos 3\theta \cos \theta + 2 \cos 2\theta \cos \theta = 0$ (by p)
 $2 \cos \theta (\cos 3\theta + \cos 2\theta) = 0$
 $2 \cos \theta (2 \cos \frac{5\theta}{2} \cos \frac{\theta}{2}) = 0$
 $4 \cos \theta \cos \frac{5\theta}{2} \cos \frac{\theta}{2} = 0$

$\cos \theta = 0 \Rightarrow \theta = \pm \frac{\pi}{2} + 2n\pi \quad n \in \mathbb{Z}$
 $\cos \frac{5\theta}{2} = 0 \Rightarrow \frac{5\theta}{2} = \pm \frac{\pi}{2} + 2n\pi \Rightarrow \theta = \pm \frac{\pi}{5} + \frac{4n\pi}{5}$
 $\cos \frac{\theta}{2} = 0 \Rightarrow \frac{\theta}{2} = \pm \frac{\pi}{2} + 2n\pi \Rightarrow \theta = \pm \pi + 4n\pi$

$\therefore$ general solns are:
 $\theta = \pm \frac{\pi}{2} + 2n\pi, \pm \frac{\pi}{5} + \frac{4n\pi}{5}, \pm \pi + 4n\pi \quad n \in \mathbb{Z}$

$$3b) y = 2 \tan^{-1} x \quad x = \tan \frac{y}{2}$$

$$\begin{aligned} \text{Vol of revol} &= \pi \int_0^{\frac{\pi}{2}} x^2 dy \\ &= \pi \int_0^{\frac{\pi}{2}} \tan^2\left(\frac{y}{2}\right) dy = \pi \int_0^{\frac{\pi}{2}} (\sec^2 \frac{y}{2} - 1) dy \\ &= \left[\pi \left(\tan \frac{y}{2} \right) 2 - \pi y \right]_0^{\frac{\pi}{2}} = 2\pi \tan \frac{\pi}{4} - \frac{\pi}{2} \\ &= 2\pi - \frac{\pi}{2} \text{ unit}^3 \end{aligned}$$



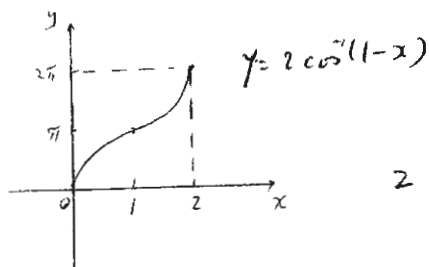
$$\begin{aligned} 4a) \int_0^{\frac{\pi}{6}} \left(\frac{1}{2} \sin 2x \right) dx &= \frac{1}{4} \int_0^{\frac{\pi}{6}} \frac{1 - \cos 4x}{2} dx \\ &= \left[\frac{x}{8} - \frac{\sin 4x}{8 \cdot 4} \right]_0^{\frac{\pi}{6}} = \frac{\pi}{48} - \frac{\sin \frac{2\pi}{3}}{32} \\ &= \frac{\pi}{48} - \frac{\sqrt{3}}{64} \end{aligned}$$

$$b) y = 1 - \cos \frac{x}{2}$$

$$\text{Period: } \frac{2\pi}{\frac{1}{2}} = 4\pi$$

i) Largest a is 2π
 ii) $f^{-1}(x)$ exists for $0 \leq x \leq 2\pi$

$$\begin{aligned} \text{ii) For inverse: } x &= 1 - \cos \frac{y}{2} \\ \cos \frac{y}{2} &= 1 - x \\ \frac{y}{2} &= \cos^{-1}(1-x) \\ y &= 2 \cos^{-1}(1-x) \end{aligned}$$



$$\begin{aligned} \text{iv) Area} &= \frac{1}{2} \times \text{Area of Rectangle} \\ &= \frac{1}{2} \times 2 \times 2\pi = 2\pi \text{ units}^2 \end{aligned}$$

Alternatively

$$\begin{aligned} \text{Area} &= 4\pi - \int_0^{\frac{\pi}{2}} (1 - \cos \frac{y}{2}) dy \\ &= 4\pi - \left[y - 2 \sin \frac{y}{2} \right]_0^{\frac{\pi}{2}} \\ &= 4\pi - (2\pi - 0 - 0 + 0) = 2\pi \text{ units}^2 \end{aligned}$$

Q5

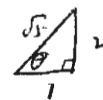
$$a) \lim_{x \rightarrow 0} \frac{x^2}{2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{x}{\sin x} \right)^2 = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$b) \text{Area } \triangle OPN = \frac{1}{2} \times 1 \times 4 \cos \theta = 2 \cos \theta$$

$$\text{Area } \triangle OPM = \frac{1}{2} \times 2 \times 4 \sin \theta = 4 \sin \theta$$

$$\text{Area OMPN(A)} = 4 \sin \theta + 2 \cos \theta$$

$$\begin{aligned} \text{ii) } A' &= 4 \cos \theta - 2 \sin \theta = 0 \quad 0 < \theta < \frac{\pi}{2} \\ 4 \cos \theta &= 2 \sin \theta \\ \tan \theta &= 2 \end{aligned}$$



$$A'' = -4 \sin \theta - 2 \cos \theta = \frac{-4 \times 2}{\sqrt{5}} - \frac{2}{\sqrt{5}} < 0$$

Since A, A', A'' are continuous and differentiable for $0 < \theta < \frac{\pi}{2}$ and only 1 TP $\therefore \tan \theta = 2$ gives max A

$$\begin{aligned} \text{iii) } P &= (4 \cos \theta, 4 \sin \theta) = \left(\frac{4}{\sqrt{5}}, \frac{4 \times 2}{\sqrt{5}} \right) \\ &= \left(\frac{4}{\sqrt{5}}, \frac{8}{\sqrt{5}} \right) \end{aligned}$$

Q6

$$\begin{aligned} \text{(i) slope } PQ &= \frac{a(q^2 - p^2)}{2a(q - p)} = \frac{a(q + p)(q - p)}{2a(q - p)} \\ &= \frac{q + p}{2} \end{aligned}$$

$$\text{Eq of PQ} \quad y - ap^2 = \left(\frac{q + p}{2} \right) (x - 2ap)$$

$$y = \frac{x}{2} (p + q) - apq - ap^2 + ap^2$$

$$y = \frac{x}{2} (p + q) - apq$$

At P and Q: $y = mx + b$ and $y = \frac{x}{2}(p+q) - apq$
 $mx + b = \frac{x}{2}(p+q) - apq$
 $m = \frac{p+q}{2} \quad b = -apq \quad \therefore pq = \frac{-b}{a} \quad \# \quad 1$

ii) Since $p+q = 2m$, $pq = -\frac{b}{a}$
 $p^2 + q^2 = (p+q)^2 - 2pq = (2m)^2 - 2(-\frac{b}{a}) = 4m^2 + \frac{2b}{a} \quad \# \quad 1$

iii) $x = -apq(p+q) = -a(-\frac{b}{a})2m \quad \therefore x = 2mb \quad \# \quad 1$
 $y = a(2 + p^2 + pq + q^2) = a(2 + (p+q)^2 - pq) = a(2 + 4m^2 - \frac{-b}{a})$
 $y = a(2 + 4m^2 + \frac{b}{a}) = 2a + 4am^2 + b \quad \# \quad 1$

iv) Since $b = \frac{2c}{2m} \quad \therefore y = 2a + 4am^2 + \frac{x}{2m}$

1 Gradient of chord PQ is fixed $\therefore m$ is a constant
 but a is also a constant $\therefore$ this is an equation of a straight line in gradient-intercept form.

$$x - 2my = -4am - 8am^3$$

1 $x + (-2m)y = 2a(-2m) + a(-2m)^3$

Compare this equation with equation of normal to parabola at 1

$$x + py = 2ap + ap^3$$

p has been replaced by $(-2m)$

$\therefore$ locus of N is a straight line which is normal to the parabola at point $(2ax - 2m, a(-2m)^2)$

1 is at $(-4m, 4am^2)$